

Лабораторная работа №16

Программный RAID

Наурузова Айшат Магомедовна

03 декабря 2025

Российский университет дружбы народов, Москва, Россия

Цель работы

Получение практических навыков создания, настройки и управления RAID-массивами в Linux (создание RAID 1, работа с hot spare, преобразование RAID 1 → RAID 5).

Ход выполнения работы

Проверка подключённых дисков

```
amnauruzova@amnauruzova:~$ su
Password:
root@amnauruzova:/home/amnauruzova#
root@amnauruzova:/home/amnauruzova# fdisk -l grep /dev/sd
fdisk: cannot open grep: No such file or directory
fdisk: cannot open /dev/sd: No such file or directory
root@amnauruzova:/home/amnauruzova# fdisk -l | grep /dev/sd
Disk /dev/sda: 1.5 GiB, 1610612736 bytes, 3145728 sectors
/dev/sda1      2048  821247  819200  400M 8e Linux LVM
/dev/sda2      821248 1435647  614400  300M 8e Linux LVM
Disk /dev/sdb: 40 GiB, 42949672960 bytes, 83886080 sectors
/dev/sdb1      2048      4095    2048    1M BIOS boot
/dev/sdb2      4096  2101247  2097152   1G Linux extended boot
/dev/sdb3  2101248 83884031 81782784  39G Linux LVM
Disk /dev/sdc: 512 MiB, 536870912 bytes, 1048576 sectors
Disk /dev/sdd: 512 MiB, 536870912 bytes, 1048576 sectors
Disk /dev/sde: 512 MiB, 536870912 bytes, 1048576 sectors
Disk /dev/sdf: 1.5 GiB, 1610612736 bytes, 3145728 sectors
/dev/sdf1      2048  616447  614400  300M 8e Linux LVM
/dev/sdf2      616448 1230847  614400  300M 8e Linux LVM
root@amnauruzova:/home/amnauruzova# sfdisk /dev/sdc <<EOF
> ;
> EOF
Checking that no-one is using this disk right now ... OK

Disk /dev/sdc: 512 MiB, 536870912 bytes, 1048576 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

>>> Created a new DOS (MBR) disklabel with disk identifier 0xb7556400.
/dev/sdc1: Created a new partition 1 of type 'Linux' and of size 511 MiB.
/dev/sdc2: Done.
```

Создание разделов на дисках

```
root@amnauruzova:/home/amnauruzova#  
root@amnauruzova:/home/amnauruzova# sfdisk --print-id /dev/sdc 1  
sfdisk: print-id is deprecated in favour of --part-type  
83  
root@amnauruzova:/home/amnauruzova# sfdisk --print-id /dev/sdd 1  
sfdisk: print-id is deprecated in favour of --part-type  
83  
root@amnauruzova:/home/amnauruzova# sfdisk --print-id /dev/sde 1  
sfdisk: print-id is deprecated in favour of --part-type  
83  
root@amnauruzova:/home/amnauruzova#  
root@amnauruzova:/home/amnauruzova# sfdisk -T grep -i raid  
sfdisk: invalid option -- 'i'  
Try 'sfdisk --help' for more information.  
root@amnauruzova:/home/amnauruzova# sfdisk -T | grep -i raid  
fd Linux raid autodetect  
root@amnauruzova:/home/amnauruzova# sfdisk --change-id /dev/sdd 1 fd  
sfdisk: change-id is deprecated in favour of --part-type  
  
The partition table has been altered.  
Calling ioctl() to re-read partition table.  
Syncing disks.  
root@amnauruzova:/home/amnauruzova# sfdisk --change-id /dev/sdc 1 fd  
sfdisk: change-id is deprecated in favour of --part-type  
  
The partition table has been altered.  
Calling ioctl() to re-read partition table.  
Syncing disks.  
root@amnauruzova:/home/amnauruzova# sfdisk --change-id /dev/sde 1 fd  
sfdisk: change-id is deprecated in favour of --part-type  
  
The partition table has been altered.  
Calling ioctl() to re-read partition table.  
Syncing disks.  
root@amnauruzova:/home/amnauruzova#
```

Проверка типа разделов

```
root@amnauruzova:/home/amnauruzova# sfdisk -l /dev/sdc
Disk /dev/sdc: 512 MiB, 536870912 bytes, 1048576 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xb7556400

Device      Boot Start      End Sectors  Size Id Type
/dev/sdc1          2048 1048575 1046528   511M fd Linux raid autodetect

root@amnauruzova:/home/amnauruzova# sfdisk -l /dev/sdd
Disk /dev/sdd: 512 MiB, 536870912 bytes, 1048576 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x60c487cc

Device      Boot Start      End Sectors  Size Id Type
/dev/sdd1          2048 1048575 1046528   511M fd Linux raid autodetect

root@amnauruzova:/home/amnauruzova# sfdisk -l /dev/sde
Disk /dev/sde: 512 MiB, 536870912 bytes, 1048576 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x1dd694ae

Device      Boot Start      End Sectors  Size Id Type
/dev/sde1          2048 1048575 1046528   511M fd Linux raid autodetect

root@amnauruzova:/home/amnauruzova#
```

Изменение ID разделов (Linux RAID)

```
root@amnauruzova:/home/amnauruzova#  
root@amnauruzova:/home/amnauruzova# mdadm --create --verbose /dev/md0 --level=1 --raid-devices=2 /dev/sdc1 /dev/sdd1  
mdadm: Note: this array has metadata at the start and  
may not be suitable as a boot device. If you plan to  
store '/boot' on this device please ensure that  
your boot-loader understands md/v1.x metadata, or use  
--metadata=0.90  
mdadm: size set to 522240K  
Continue creating array [y/N]? y  
mdadm: Defaulting to version 1.2 metadata  
mdadm: array /dev/md0 started.  
root@amnauruzova:/home/amnauruzova# cat /proc/mdstat  
Personalities : [raid1]  
md0 : active raid1 sdd1[1] sdc1[0]  
522240 blocks super 1.2 [2/2] [UU]  
  
unused devices: <none>  
root@amnauruzova:/home/amnauruzova# mdadm --query /dev/md0  
/dev/md0: 510.00MiB raid1 2 devices, 0 spares. Use mdadm --detail for more detail.  
root@amnauruzova:/home/amnauruzova#
```

Рис. 4: Изменение ID

Создание массива RAID 1

```
root@amnauruzova:/home/amnauruzova# mdadm --detail /dev/md0
/dev/md0:
    Version : 1.2
  Creation Time : Tue Nov 18 15:47:52 2025
    Raid Level : raid1
    Array Size : 522240 (510.00 MiB 534.77 MB)
  Used Dev Size : 522240 (510.00 MiB 534.77 MB)
    Raid Devices : 2
    Total Devices : 2
 Persistence : Superblock is persistent

    Update Time : Tue Nov 18 15:47:54 2025
      State : clean
    Active Devices : 2
   Working Devices : 2
   Failed Devices : 0
    Spare Devices : 0

Consistency Policy : resync

    Name : amnauruzova.localdomain:0 (local to host amnauruzova.localdomain)
    UUID : 770a4376:13313565:be9a28f1:e5098b0e
    Events : 17

   Number Major Minor RaidDevice State
     0       8      33         0   active sync  /dev/sdc1
     1       8      49         1   active sync  /dev/sdd1
root@amnauruzova:/home/amnauruzova#
```

Рис. 5: Создание RAID

Проверка состояния RAID 1

```
GNU nano 8.1 /etc/fstab

#
# /etc/fstab
# Created by anaconda on Tue Sep  2 12:53:30 2025
#
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
#
# After editing this file, run 'systemctl daemon-reload' to update systemd
# units generated from this file.
#
UUID=34884054-27f2-4624-8671-1f66bf893ae2 / xfs defaults 0 0
UUID=5bb25999-de96-4a24-930c-75854bf11fc1 /boot xfs defaults 0 0
UUID=774abe3d-c83f-4c4f-90c6-404f43a900ee none swap defaults 0 0
/dev/vgdata/lvdata /mnt/data ext4 defaults 1 2
/dev/vggroup/lvggroup /mnt/groups xfs defaults 1 2
/dev/md0 /data/raid ext4 defaults 1 2

#UUID=e97bbe69-dd87-4579-880c-1ca10bbb317 /mnt/data xfs defaults 1 2
#UUID=ef6eb4a8-920f-4d8b-96f8-b4e9a925c30b /mnt/data-ext ext4 defaults 1 2
#UUID=6c12fcce-02a4-4084-8214-c952a4d0cbda none swap defaults 0 0
```

Рис. 6: Проверка RAID

Настройка монтирования и запись в fstab

```
root@amnauruzova:/home/amnauruzova# mdadm /dev/md0 --fail /dev/sdc1
root@amnauruzova:/home/amnauruzova# mdadm /dev/md0 --remove /dev/sdc1
mdadm: hot removed /dev/sdc1 from /dev/md0
root@amnauruzova:/home/amnauruzova# mdadm /dev/md0 --add /dev/sde1
mdadm: added /dev/sde1
root@amnauruzova:/home/amnauruzova# mdadm --detail /dev/md0
/dev/md0:
    Version : 1.2
  Creation Time : Tue Nov 18 15:47:52 2025
    Raid Level : raid1
    Array Size : 522240 (510.00 MiB 534.77 MB)
  Used Dev Size : 522240 (510.00 MiB 534.77 MB)
    Raid Devices : 2
    Total Devices : 2
     Persistence : Superblock is persistent

    Update Time : Tue Nov 18 15:51:05 2025
      State : clean
    Active Devices : 2
    Working Devices : 2
    Failed Devices : 0
    Spare Devices : 0

Consistency Policy : resync

    Name : amnauruzova.localdomain:0 (local to host amnauruzova.localdomain)
    UUID : 770a4376:13313565:be9a28f1:e5098b0e
    Events : 39

   Number Major Minor RaidDevice State
     2       8      65          0  active sync  /dev/sde1
     1       8      49          1  active sync  /dev/sdd1
root@amnauruzova:/home/amnauruzova#
```

```
root@amnauruzova:/home/amnauruzova#  
root@amnauruzova:/home/amnauruzova# umount /dev/md0  
root@amnauruzova:/home/amnauruzova# mdadm --stop /dev/md0  
mdadm: stopped /dev/md0  
root@amnauruzova:/home/amnauruzova# mdadm --zero-superblock /dev/sdc1  
root@amnauruzova:/home/amnauruzova# mdadm --zero-superblock /dev/sdd1  
root@amnauruzova:/home/amnauruzova# mdadm --zero-superblock /dev/sde1  
root@amnauruzova:/home/amnauruzova# █
```

Рис. 8: Сбой диска

Создание RAID 1 и добавление резервного диска

```
root@amnauruzova:/home/amnauruzova# mdadm --create --verbose /dev/md0 --level=1 --raid-devices=2 /dev/sdc1 /dev/sdd1
mdadm: Note: this array has metadata at the start and
may not be suitable as a boot device.  If you plan to
store '/boot' on this device please ensure that
your boot-loader understands md/v1.x metadata, or use
--metadata=0.90
mdadm: size set to 522240K
Continue creating array [y/N]? y
mdadm: Defaulting to version 1.2 metadata
mdadm: array /dev/md0 started.
root@amnauruzova:/home/amnauruzova# mdadm --add /dev/md0 /dev/sde1
mdadm: added /dev/sde1
root@amnauruzova:/home/amnauruzova# mount /dev/md0
mount: (hint) your fstab has been modified, but systemd still uses
the old version; use 'systemctl daemon-reload' to reload.
root@amnauruzova:/home/amnauruzova# cat /proc/mdstat
Personalities : [raid1]
md0 : active raid1 sde1[2](S) sdd1[1] sdc1[0]
      522240 blocks super 1.2 [2/2] [UU]

unused devices: <none>
root@amnauruzova:/home/amnauruzova# mdadm --query /dev/md0
/dev/md0: 510.00MiB raid1 2 devices, 1 spare. Use mdadm --detail for more detail.
root@amnauruzova:/home/amnauruzova#
```

Рис. 9: RAID hot spare

Проверка состояния массива

```
root@amnauruzova:/home/amnauruzova# mdadm --detail /dev/md0
/dev/md0:
    Version : 1.2
  Creation Time : Tue Nov 18 15:53:02 2025
    Raid Level : raid1
    Array Size : 522240 (510.00 MiB 534.77 MB)
  Used Dev Size : 522240 (510.00 MiB 534.77 MB)
    Raid Devices : 2
  Total Devices : 3
 Persistence : Superblock is persistent

    Update Time : Tue Nov 18 15:53:24 2025
      State : clean
  Active Devices : 2
 Working Devices : 3
 Failed Devices : 0
  Spare Devices : 1


Consistency Policy : resync


    Name : amnauruzova.localdomain:0 (local to host amnauruzova.localdomain)
    UUID : 272a859d:84981356:fd01ec6a:d58e8d83
    Events : 18


   Number Major Minor RaidDevice State
     0       8      33        0   active sync  /dev/sdc1
     1       8      49        1   active sync  /dev/sdd1

     2       8      65        -   spare   /dev/sde1

root@amnauruzova:/home/amnauruzova#
```

Имитация сбоя и автоматическое восстановление

```
root@amnauruzova:/home/amnauruzova# mdadm /dev/md0 --fail /dev/sdc1
root@amnauruzova:/home/amnauruzova# mdadm --detail /dev/md0
/dev/md0:
    Version : 1.2
  Creation Time : Tue Nov 18 15:53:02 2025
    Raid Level : raid1
    Array Size : 522240 (510.00 MiB 534.77 MB)
  Used Dev Size : 522240 (510.00 MiB 534.77 MB)
    Raid Devices : 2
    Total Devices : 3
    Persistence : Superblock is persistent

    Update Time : Tue Nov 18 15:54:35 2025
      State : clean
    Active Devices : 2
    Working Devices : 2
    Failed Devices : 1
    Spare Devices : 0

Consistency Policy : resync

    Name : amnauruzova.localdomain:0 (local to host amnauruzova.localdomain)
    UUID : 272a859d:84981356:fd01ec6a:d58e8d83
    Events : 37

   Number  Major   Minor  RaidDevice State
     2         8       65           0  active sync  /dev/sde1
     1         8       49           1  active sync  /dev/sdd1

     0         8       33           -  faulty      /dev/sdc1

root@amnauruzova:/home/amnauruzova#
```

Исходное состояние RAID 1

```
root@amnauruzova:/home/amnauruzova# mdadm --create --verbose /dev/md0 --level=1 --raid-devices=2 /dev/sdc1 /dev/sdd1
mdadm: Note: this array has metadata at the start and
may not be suitable as a boot device. If you plan to
store '/boot' on this device please ensure that
your boot-loader understands md/v1.x metadata, or use
--metadata=0.90
mdadm: size set to 522240K
Continue creating array [y/N]? y
mdadm: Defaulting to version 1.2 metadata
mdadm: array /dev/md0 started.
root@amnauruzova:/home/amnauruzova# mdadm --add /dev/md0 /dev/sde1
mdadm: added /dev/sde1
root@amnauruzova:/home/amnauruzova# mount /dev/md0
mount: (hint) your fstab has been modified, but systemd still uses
the old version; use 'systemctl daemon-reload' to reload.
root@amnauruzova:/home/amnauruzova# cat /proc/mdstat
Personalities : [raid1]
md0 : active raid1 sde1[2](S) sdd1[1] sdc1[0]
      522240 blocks super 1.2 [2/2] [UU]

unused devices: <none>
root@amnauruzova:/home/amnauruzova# mdadm --query /dev/md0
/dev/md0: 510.00MiB raid1 2 devices, 1 spare. Use mdadm --detail for more detail.
root@amnauruzova:/home/amnauruzova#
```

Рис. 12: Исходный RAID1

Проверка состояния перед изменением уровня

```
root@amnauruzova:/home/amnauruzova# mdadm --detail /dev/md0
/dev/md0:
    Version : 1.2
  Creation Time : Tue Nov 18 15:58:47 2025
    Raid Level : raid1
    Array Size : 522240 (510.00 MiB 534.77 MB)
  Used Dev Size : 522240 (510.00 MiB 534.77 MB)
    Raid Devices : 2
  Total Devices : 3
 Persistence : Superblock is persistent

    Update Time : Tue Nov 18 15:59:13 2025
      State : clean
 Active Devices : 2
Working Devices : 3
 Failed Devices : 0
  Spare Devices : 1


Consistency Policy : resync

    Name : amnauruzova.localdomain:0 (local to host amnauruzova.localdomain)
   UUID : 69767fbb:b2e3b315:20c82e3b:c30d0b5d
  Events : 18

   Number Major Minor RaidDevice State
    0       8     33        0  active sync  /dev/sdc1
    1       8     49        1  active sync  /dev/sdd1

    2       8     65         -  spare   /dev/sde1
root@amnauruzova:/home/amnauruzova#
```

Рис. 13: Перед изменением

Преобразование в RAID 5

```
root@amnauruzova:/home/amnauruzova# mdadm --grow /dev/md0 --level=5
mdadm: level of /dev/md0 changed to raid5
root@amnauruzova:/home/amnauruzova# mdadm --detail /dev/md0
/dev/md0:
    Version : 1.2
  Creation Time : Tue Nov 18 15:58:47 2025
    Raid Level : raid5
    Array Size : 522240 (510.00 MiB 534.77 MB)
  Used Dev Size : 522240 (510.00 MiB 534.77 MB)
    Raid Devices : 2
    Total Devices : 3
 Persistence : Superblock is persistent

    Update Time : Tue Nov 18 16:00:21 2025
      State : clean
 Active Devices : 2
Working Devices : 3
 Failed Devices : 0
 Spare Devices : 1


    Layout : left-symmetric
    Chunk Size : 64K

Consistency Policy : resync

           Name : amnauruzova.localdomain:0 (local to host amnauruzova.localdomain)
          UUID : 69767fbb:b2e3b315:20c82e3b:c30d0b5d
        Events : 19

   Number   Major   Minor   RaidDevice State
     0       8       33         0     active sync   /dev/sdc1
     1       8       49         1     active sync   /dev/sdd1

     2       8       65         -     spare    /dev/sde1
root@amnauruzova:/home/amnauruzova#
```

Масштабирование RAID 5 до трёх дисков

```
root@amnauruzova:/home/amnauruzova# mdadm --grow /dev/md0 --raid-devices=3
root@amnauruzova:/home/amnauruzova# mdadm --detail /dev/md0
/dev/md0:
    Version : 1.2
    Creation Time : Tue Nov 18 15:58:47 2025
    Raid Level : raid5
    Array Size : 1044480 (1020.00 MiB 1069.55 MB)
    Used Dev Size : 522240 (510.00 MiB 534.77 MB)
    Raid Devices : 3
    Total Devices : 3
    Persistence : Superblock is persistent

    Update Time : Tue Nov 18 16:01:01 2025
    State : clean
    Active Devices : 3
    Working Devices : 3
    Failed Devices : 0
    Spare Devices : 0


    Layout : left-symmetric
    Chunk Size : 64K

Consistency Policy : resync

    Name : amnauruzova.localdomain:0 (local to host amnauruzova.localdomain)
    UUID : 69767fbb:b2e3b315:20c82e3b:c30d0b5d
    Events : 37

    Number Major Minor RaidDevice State
    0        8      33        0     active sync  /dev/sdc1
    1        8      49        1     active sync  /dev/sdd1
    2        8      65        2     active sync  /dev/sde1

root@amnauruzova:/home/amnauruzova#
```

Итоги работы

В ходе выполнения работы были созданы и настроены массивы RAID 1 и RAID 5, изучены принципы их функционирования, восстановление после отказов и возможности изменения уровня массива.

Была рассмотрена концепция горячего резерва и выполнено масштабирование RAID. Полученные навыки позволяют эффективно управлять программными RAID-массива в Linux.